Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

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EXPERIMENT 18

Understand the KNN algorithm for a given dataset

Experiment 18: Understand the KNN algorithm for a given dataset

Aim

To apply the K-nearest neighbors algorithm to the given dataset.

Source Code

```
import numpy as np
import pandas as pd
df=pd.read_csv('Iris.csv')
df.info()
df.variety.value_counts()
df.head()
features=df.iloc[:,:-1].values
label=df.iloc[:,4].values
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,rando
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))
from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

Output - Experiment 18

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
Setosa 50
Versicolor 50
Virginica 50
Name: variety, dtype: int64
sepal.length sepal.width petal.length petal.width variety
0 5.1 3.5 1.4 0.2 Setosa
1 4.9 3.0 1.4 0.2 Setosa
2 4.7 3.2 1.3 0.2 Setosa
3 4.6 3.1 1.5 0.2 Setosa
4 5.0 3.6 1.4 0.2 Setosa
KNeighborsClassifier()
0.9583333333333334
1.0
array([[50, 0, 0],
[0, 47, 3],
[0, 2, 48]], dtype=int64)
precision recall f1-score support
Setosa 1.00 1.00 1.00 50
Versicolor 0.96 0.94 0.95 50
Virginica 0.94 0.96 0.95 50
accuracy 0.97 150
macro avg 0.97 0.97 0.97 150
weighted avg 0.97 0.97 0.97 150

Result - Experiment 18

The K-nearest neighbors algorithm was applied to the given dataset.